

$$\begin{array}{ccccc}
 & & & & \\
 & & & & \\
 & & & & \\
 v_i \in & V & \xrightarrow{T} & W & \ni Tv_i \\
 & \uparrow \Phi_B^{-1} & & \downarrow \Phi_C & \\
 e_i \in & K^n & \xrightarrow{\Phi_C \circ T \circ \Phi_B^{-1}} & K^m & \ni [Tv_i]_C
 \end{array}$$